

Q1 JEE Main 2020 - 2 September (Morning)

For octahedral Mn(II) and tetrahedral Ni(II) complexes, consider the following statements:

- (I) both the complexes can be high spin.
- (II) Ni(II) complex can very rarely be low spin.
- (III) with strong field ligands, Mn(II) complexes can be low spin.
- (IV) aqueous solution of Mn(II) ions is yellow in colour.

The correct statements are :

- (A) (I), (III) and (IV) only
- (B) (I) and (II) only
- (C) (II), (III) and (IV) only
- (D) (I), (II) and (III) only

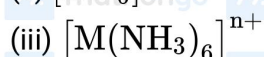
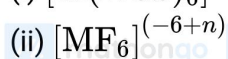
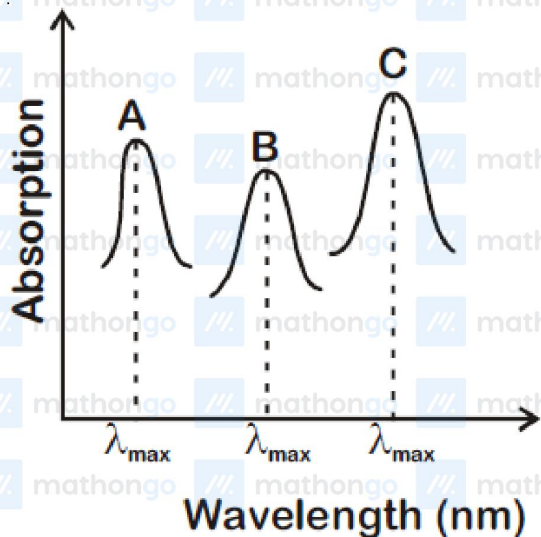
Q2 JEE Main 2020 - 2 September (Morning)

Consider that a d^6 metal ion (M^{2+}) forms a complex with aqua ligands, and the spin only magnetic moment of the complex is 4.90 BM. The geometry and the crystal field stabilization energy of the complex is

- (A) tetrahedral and $-1.6\Delta_t + 1P$
- (B) octahedral and $-2.4\Delta_0 + 2P$
- (C) tetrahedral and $-0.6\Delta_t$
- (D) octahedral and $-1.6\Delta_0$

Q3 JEE Main 2020 - 2 September (Evening)

Simplified absorption spectra of three complexes ((i), (ii) and (iii)) of Mn^{n+} ion are provided below; their $\lambda_{\max}$ values are marked as A, B and C respectively. The correct match between the complexes and their $\lambda_{\max}$ values is



(A) A – (i), B – (ii), C – (iii)

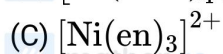
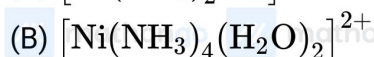
(B) A – (ii), B – (iii), C – (i)

(C) A – (ii), B – (i), C – (iii)

(D) A – (iii), B – (i), C – (ii)

Q4 JEE Main 2020 - 2 September (Evening)

The one that is not expected to show isomerism is



Q5 JEE Main 2020 - 3 September (Morning)

The electronic spectrum of $[Ti(H_2O)_6]^{3+}$ shows a single broad peak with a maximum at $20,300\text{cm}^{-1}$. The crystal field stabilization energy (CFSE) of the complex ion, in kJ mol^{-1} , is

#MathBoleTohMathonGo

www.mathongo.com

Coordination Compounds

Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

$$(1\text{kJmol}^{-1} = 83.7\text{cm}^{-1})$$

- (A) 145.5
- (B) 97
- (C) 242.5
- (D) 83.7

Q6 JEE Main 2020 - 3 September (Morning)

The complex that can show optical activity

- (A) $\text{cis} [\text{Fe}(\text{NH}_3)_2(\text{CN})_4]^-$
- (B) $\text{trans} - [\text{Cr}(\text{Cl}_2)(\text{ox})_2]^{3-}$
- (C) $\text{trans} - [\text{Fe}(\text{NH}_3)_2(\text{CN})_4]$
- (D) $\text{cis} - [\text{CrCl}_2(\text{ox})_2]^{3-}$ (ox = oxalate)

Q7 JEE Main 2020 - 3 September (Evening)

The d-electron configuration of $[\text{Ru}(\text{en})_3] \text{Cl}_2$ and $[\text{Fe}(\text{H}_2\text{O})_6] \text{Cl}_2$, respectively are

- (A) $t_{2g}^6 e_g^0$ and $t_{2g}^6 e_g^0$
- (B) $t_{2g}^6 e_g^0$ and $t_{2g}^4 e_g^2$
- (C) $t_{2g}^4 e_g^2$ and $t_{2g}^4 e_g^2$
- (D) $t_{2g}^4 e_g^2$ and $t_{2g}^6 e_g^0$

Q8 JEE Main 2020 - 3 September (Evening)

Complex A has a composition of $\text{H}_{12}\text{O}_6\text{Cl}_3\text{Cr}$. If the complex on treatment with conc. H_2SO_4 loses 13.5% of its original mass, the correct molecular formula of A is
[Given : atomic mass of Cr = 52 amu and Cl = 35amu]

- (A) $[\text{Cr}(\text{H}_2\text{O})_6] \text{Cl}_3$
- (B) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2] \text{Cl} \cdot 2\text{H}_2\text{O}$
- (C) $[\text{Cr}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$
- (D) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}] \text{Cl}_2 \cdot \text{H}_2\text{O}$

Coordination Compounds

Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

Q9 JEE Main 2020 - 4 September (Morning)

The number of isomers possible for $[\text{Pt}(\text{en})(\text{NO}_2)_2]$ is

- (A) 3
- (B) 1
- (C) 4
- (D) 2

Q10 JEE Main 2020 - 4 September (Morning)

The pair in which both the species have the same magnetic moment (spin only) is

- (A) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{CoCl}_4]^{2-}$
- (B) $[\text{Co}(\text{OH})_4]^{2-}$ and $[\text{Fe}(\text{NH}_3)_6]^{2+}$
- (C) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$
- (D) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$

Q11 JEE Main 2020 - 4 September (Evening)

The one that can exhibit highest paramagnetic behaviour among the following is
gly = glycinate; bpy = 2, 2'-bipyridine

- (A) $[\text{Fe}(\text{en})(\text{bpy})(\text{NH}_3)_2]^{2+}$
- (B) $[\text{Pd}(\text{gly})_2]$
- (C) $[\text{Co}(\text{OX})_2(\text{OH})_2]^- (\Delta_0 > P)$
- (D) $[\text{Ti}(\text{NH}_3)_6]^{3+}$

Q12 JEE Main 2020 - 4 September (Evening)

The Crystal Field Stabilization Energy (CFSE) of $[\text{CoF}_3(\text{H}_2\text{O})_3] (\Delta_0 < P)$ is

- (A) $-0.8 \Delta_0$
- (B) $-0.8 \Delta_0 + 2P$
- (C) $-0.4 \Delta_0$
- (D) $-0.4 \Delta_0 + P$

Q13 JEE Main 2020 - 4 September (Evening)

The molecule in which hybrid MOs involve only one d-orbital of the central atom is

- (A) XeF_4
- (B) $[\text{Ni}(\text{CN})_4]^{2-}$
- (C) $[\text{CrF}_6]^{3-}$
- (D) BrF_5

Q14 JEE Main 2020 - 5 September (Morning)

The values of the crystal field stabilization energies for a high spin d^6 metal ion in octahedral and tetrahedral fields, respectively,

- (A) $-1.6\Delta_0$ and $-0.4\Delta_t$
- (B) $-2.4\Delta_0$ and $-0.64\Delta_t$
- (C) $-0.4\Delta_0$ and $-0.27\Delta_t$
- (D) $-0.4\Delta_0$ and $-0.6\Delta_t$

Q15 JEE Main 2020 - 5 September (Morning)

The total number of coordination sites in ethylenediaminetetraacetate (EDTA^{4-}) is

Q16 JEE Main 2020 - 5 September (Evening)

Consider the complex ions,

trans- $[\text{Co}(\text{en})_2\text{Cl}_2]^+$ (A) and cis- $[\text{Co}(\text{en})_2\text{Cl}_2]^+$ (B)

The correct statement regarding them is

- (A) both (A) and (B) can be optically active.
- (B) both (A) and (B) can not be optically active.
- (C) (A) can not be optically active, but (B) can be optically active.
- (D) (A) can be optically active, but (B) can not be optically active.

Q17 JEE Main 2020 - 5 September (Evening)

Considering that $\Delta_0 > P$, the magnetic moment (in BM) of $[\text{Ru}(\text{H}_2\text{O})_6]^{2+}$ would be

Q18 JEE Main 2020 - 6 September (Morning)

The species that has a spin-only magnetic moment of 5.9 BM, is (T_d = tetrahedral)

- (A) $[\text{MnBr}_4]^{2-}$ (T_d)
- (B) $[\text{NiCl}_4]^{2-}$ (T_d)
- (C) $\text{Ni}(\text{CO})_4$ (T_d)
- (D) $[\text{Ni}(\text{CN})_4]^{2-}$ (square planar)

Q19 JEE Main 2020 - 6 September (Evening)

For a d metal ion in an octahedral field, the correct electronic configuration is

- (A) $t_{2g}^4 e_g^0$ when $\Delta_0 < P$
- (B) $t_{2g}^3 e_g^1$ when $\Delta_0 > P$
- (C) $e_g^2 t_{2g}^2$ when $\Delta_0 < P$
- (D) $t_{2g}^3 e_g^1$ when $\Delta_0 < P$

Q20 JEE Main 2020 - 7 January (Morning)

Correct IUPAC name of $[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NH}_2\text{CH}_3)] \text{Cl}$ is:

- (A) Bis(amine)chlorido(methylamine)platinum(II) chloride
- (B) Diamminechlorido(methanamine)platinum(II)chloride
- (C) Diamminechlorido(aminomethyl)platinum(II) chloride
- (D) Chloridobis(amine)methylamineplatinum(II) chloride

Q21 JEE Main 2020 - 7 January (Evening)

Which of the following statements are incorrect ?

- (A) Co^{+3} with strong field ligand forms high magnetic moment complex.
- (B) For Co^{+3} if pairing energy (P) $> \Delta_0$ then the complex formed will have $t_{2g}^4 e_g^2$ configuration
- (C) For $[\text{Co}(\text{en})_3]^{3+}$ $\lambda_{\text{absorbed}}$ is less than $\lambda_{\text{absorbed}}$ for $[\text{CoF}_6]^{3-}$
- (D) If $\Delta_0 = 18000 \text{ cm}^{-1}$ for Co^{+3} then with same ligands for it $\Delta_t = 16000 \text{ cm}^{-1}$

- (A) A, D

Coordination Compounds

Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

(B) B, C

(C) A, B

(D) A

Q22 JEE Main 2020 - 7 January (Evening)

For the complex $[Ma_2b_2]$ if M is sp^3 or dsp^2 hybridised respectively then total number of optical isomers are respectively :

(A) 1, 1

(B) 2, 1

(C) 0, 0

(D) 1, 2

Q23 JEE Main 2020 - 8 January (Morning)

Which of the following complex exhibit facial meridional geometrical isomerism.

(A) $[Pt(NH_3)Cl_3]^-$

(B) $[PtCl_2(NH_3)_2]$

(C) $[Ni(CO)_4]$

(D) $[Co(NO_2)_3(NH_3)_3]$

Q24 JEE Main 2020 - 8 January (Morning)

The volume (in mL) of $0.125 M AgNO_3$ required to quantitatively precipitate chloride ions in $0.3 g$ of $[Co(NH_3)_6]Cl_3$ is

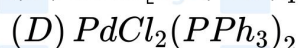
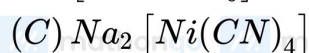
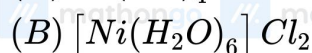
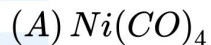
$$^M[Co(NH_3)_6]Cl_3 = 267.46 g/mol$$

$$^M[AgNO_3] = 169.87 g/mol$$

Multiply your answer with 100

Q25 JEE Main 2020 - 8 January (Evening)

The correct order of the calculated spin-only magnetic moments of complexes (A) to (D) is



(A) $a = c = d < b$

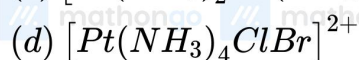
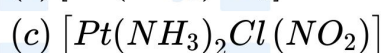
(B) $a < b < c < d$

(C) $a > b > c > d$

(D) $a = b > c > d$

Q26 JEE Main 2020 - 8 January (Evening)

Among (a)-(d), the complexes that can display geometrical isomerism are :



(A) c, d

(B) a, b

(C) a, b & c

(D) a, b, c & d

Q27 JEE Main 2020 - 8 January (Evening)

Complexes (ML_5) of metal Ni and Fe have ideal square pyramidal and trigonal bipyramidal geometries, respectively. The sum of the 90° , 120° and 180° $L-M$ angles in the two complexes is . . .

Q28 JEE Main 2020 - 9 January (Morning)

Complex X of composition of $Cr(H_2O)_6Cl_n$ has a spin only magnetic moment of 3.83 BM . It reacts with $AgNO_3$ and shows geometrical isomerism. The IUPAC nomenclature of X is :

(A) Hexaaquachromium (III) chloride

Coordination Compounds

Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

- (B) Tetraaquadichloridochromium (III) chloride dihydrate
(C) Dichloridotetraaqua chromium(IV) chloride dehydrate
(D) Tetraaquadichloridochromium (IV) chloride dehydrate

Q29 JEE Main 2020 - 9 January (Morning)

$[Pd(F)(Cl)(Br)(I)]^{2-}$ has n number of geometrical isomers. Then, the spin-only magnetic moment and crystal field stabilisation energy $[CFSE]$ of $[Fe(CN)_6]^{n-6}$, respectively, are :
[Note : Ignore the pairing energy]

- (A) 1.73 B. M. and $-2\Delta_0$
(B) 2.84 B. M. and $-1.6\Delta_0$
(C) 0 and $-1.6\Delta_0$
(D) 5.92 B. M. and $-2.4\Delta_0$

Q30 JEE Main 2020 - 2 September (Morning)

The oxidation states of iron atoms in compounds (A), (B) and (C), respectively, are x , y and z . The sum of x , y and z is



Coordination Compounds

Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

Answer Key

Q1 (D)

Q5 (B)

Q9 (A)

Q13 (B)

Q17 (O)

Q21 (A)

Q25 (A)

Q29 (A)

Q2 (C)

Q6 (D)

Q10 (D)

Q14 (D)

Q18 (A)

Q22 (C)

Q26 (A)

Q30 (6)

Q3 (A)

Q7 (B)

Q11 (C)

Q15 (6)

Q19 (D)

Q23 (D)

Q27 (20)

Q4 (B)

Q8 (B)

Q12 (C)

Q16 (C)

Q20 (B)

Q24 (2692)

Q28 (B)

#MathBoleTohMathonGo

www.mathongo.com

Coordination Compounds

Hints & Solutions

JEE Main 2020 Chapterwise

Chemistry

Q1

- Both complex can be high spin
- Tetrahedral complex are generally high spin complex with any ligand is present
- Aqueous solution of Mn^{2+} ions is pink in colour
- $\text{Mn}^{2+} + \text{SFL} \rightarrow \text{low spin complex}$

Q2

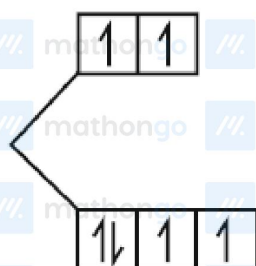
Spin only magnetic moment $= 4.9 = \sqrt{n(n+2)}$

From this, $n = 4$ (unpaired electrons)

In case of d^6 having 4 unpaired electrons.

We have 2 possibilities.

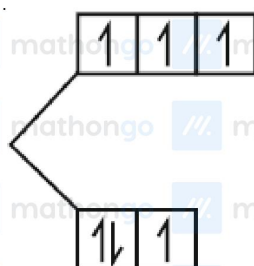
For octahedral



$$\text{CFSE} = -0.4 \Delta_0$$

So, option (3) is correct.

For tetrahedral



$$\text{CFSE} = -0.6 \Delta_t$$

Q3

$$(\lambda_{\max})_C > (\lambda_{\max})_B > (\lambda_{\max})_A$$

From spectrochemical series



$\Rightarrow A - iii, B - i, C - ii$

Q4

#MathBoleTohMathonGo

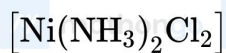
www.mathongo.com

Coordination Compounds

Hints & Solutions

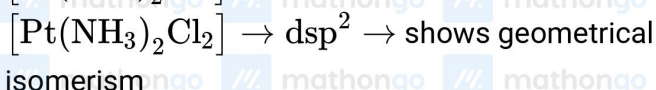
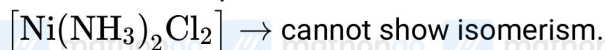
JEE Main 2020 Chapterwise

Chemistry



$\text{Ni}^{2+} \rightarrow d^8$ sp^3 hybridisation

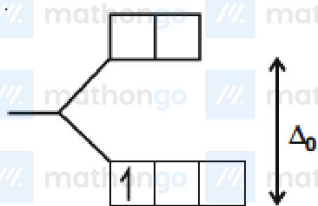
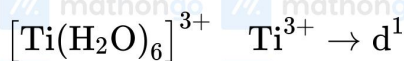
Tetrahedral complex



$[\text{Ni}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+} \rightarrow \text{Octahedral} \rightarrow \text{show geometrical isomerism}$

$[\text{Ni}(\text{en})_3]^{2+} \rightarrow \text{Octahedral} \rightarrow \text{shows optical isomerism}$

Q5



$$\Delta_0 = 20300 \text{ cm}^{-1}$$

For octahedral

$$\text{CFSE} = -0.4\Delta_0 = -0.4 \times 20300 = -8120 \text{ cm}^{-1}$$

$$\text{CFSE (in kJ)} = \frac{8120}{83.7} = 97 \text{ kJ/mol}$$

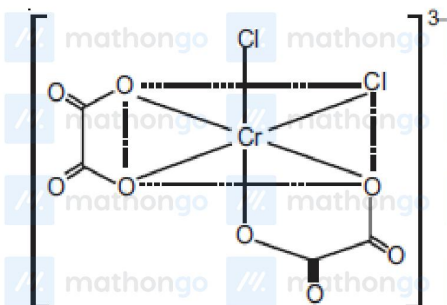
Q6

Coordination Compounds

Hints & Solutions

JEE Main 2020 Chapterwise

Chemistry

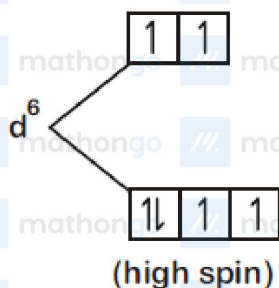
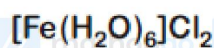
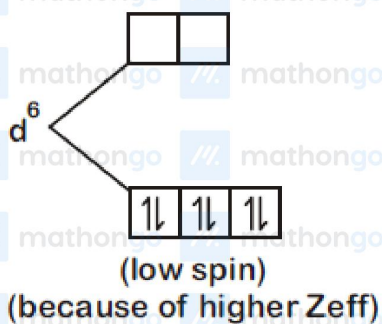


does not have symmetry, so, optically active.

$\{cis - [\text{Fe}(\text{NH}_3)_2(\text{CN})_4]^{-}, trans [\text{CrCl}_2(\text{Ox})_2]^{3-}, trans - [\text{Fe}(\text{NH}_3)_2(\text{CN})_4]^{-}\}$ have plane of symmetry.

So, optically inactive.

Q7



Coordination Compounds

Hints & Solutions

JEE Main 2020 Chapterwise

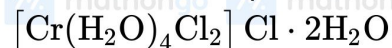
Chemistry

Q8

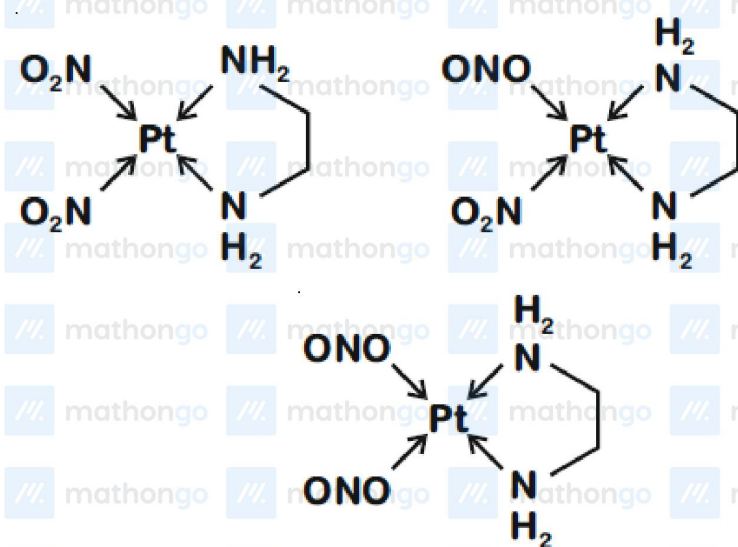
13.5% of $\text{H}_{12}\text{O}_6\text{Cl}_3\text{Cr}$ is equal to 36 (approx.)

$\therefore$ around two moles of water are lost during heating.

$\therefore$ Formula of complex could be



Q9

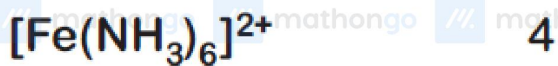
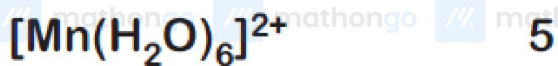
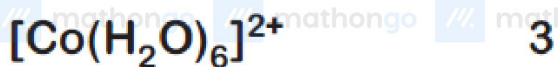
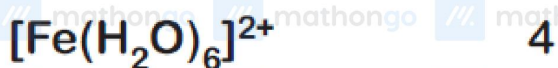
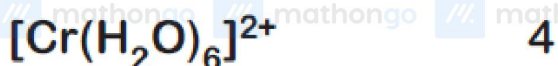


Total 3 geometrical isomers are possible.

Q10

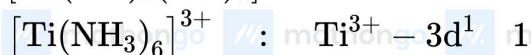
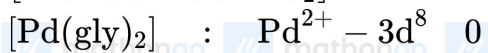
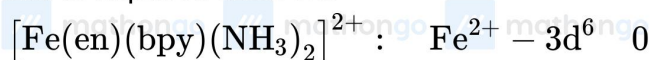
Species with same number of unpaired electrons have equal magnetic moment

Complex (n)



Q11

No. of unpaired electrons



$\therefore [\text{Co}(\text{OX})_2(\text{OH})_2]^-$
exhibits highest paramagnetic behaviour.

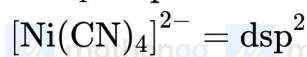
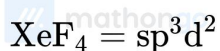
Q12



$$\text{CFSE} = [-4 \times 0.4 + 2 \times 0.6] \Delta_0 + 0$$

$$= -0.4 \Delta_0$$

Q13

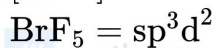
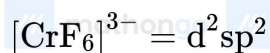


Coordination Compounds

Hints & Solutions

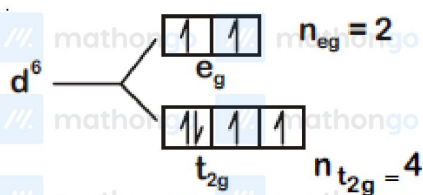
JEE Main 2020 Chapterwise

Chemistry



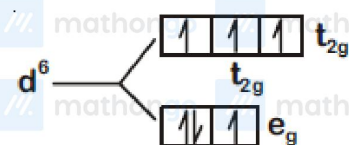
Q14

Crystal field stabilization energy (CFSE) for high spin d^6 metal ion
In octahedral



$$\begin{aligned}\text{CFSE} &= [-0.4n_{t_{2g}} + 0.6n_{e_g}] \Delta_0 \\ &= [-0.4 \times 4 + 0.6 \times 2] \Delta_0 \\ &= -0.4\Delta_0\end{aligned}$$

In tetrahedral fields



$$\begin{aligned}\text{CFSE} &= (-0.6n_{e_g} + 0.4n_{t_{2g}}) \Delta_t \\ &= (-0.6 \times 3 + 0.4 \times 3) \Delta_t \\ &= (-1.8 + 1.2) \Delta_t \\ &= -0.6\Delta_t\end{aligned}$$

Q15

$[\text{EDTA}]^{4-}$ is ethylenediaminetetraacetate anion.

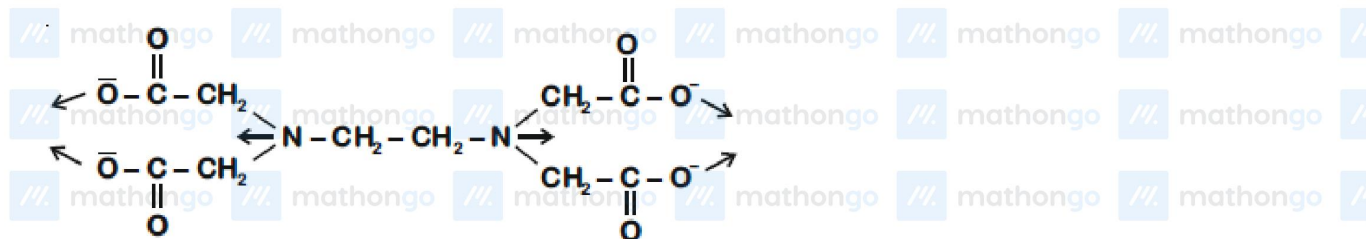
It is a hexadentate ligand.

Coordination Compounds

Hints & Solutions

JEE Main 2020 Chapterwise

Chemistry



It has six co-ordination sites.

Q16

trans-[Co(en)₂Cl₂]⁺ (A) contain a plane of symmetry. It cannot be optically active. Whereas, (B) cis-[Co(en)₂Cl₂]⁺ does not contain any plane of symmetry. The compound (B) is optically active.

Q17

As $\Delta_0 > P$

$\therefore$ Pairing of e⁻s will take place

Ru²⁺ (Z = 44) = [Kr]4d⁶

No. of unpaired e⁻s = 0

$\therefore$ Magnetic moment = 0B.M

Q18

[MnBr₄]²⁻ $\Rightarrow$ Mn²⁺ $\Rightarrow$ d⁵(Td)

d⁵(Td) is high spin complex

$\mu = \sqrt{5(5+2)} = 5.91\text{B} \cdot \text{M}$

Q19

If $\Delta_0 > P$ then for d⁴ $\Rightarrow$ t_{2g}⁴e_g⁰

If $\Delta_0 < P$ then for d⁴ $\Rightarrow$ t_{2g}³e_g¹

Q20

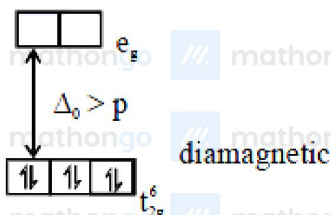
#MathBoleTohMathonGo

www.mathongo.com

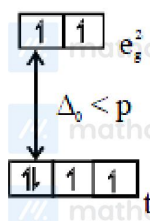
Self-Explanatory

Q21

(a) Co^{+3} (with strong field ligands)



(b) If $\Delta_o < p$:



(c) Splitting power of ethylenediamine (en) is greater than fluoride (F^-) ligand therefore more energy absorbed by $[\text{Co}(\text{en})_3]^{3+}$ as compared to $[\text{CoF}_6]^{3-}$. So wavelength of light is absorbed by $[\text{Co}(\text{en})_3]^{3+}$ is lower than that of $[\text{CoF}_6]^{3-}$.

(D)

$$\Delta_t = \frac{4}{9} \Delta_o$$

so if $\Delta_o = 18000 \text{ cm}^{-1}$

$$\Delta_t = \frac{4}{9} \times 18000 = 8000 \text{ cm}^{-1}$$

Q22

Both will not show optical isomerism.

Coordination Compounds

Hints & Solutions

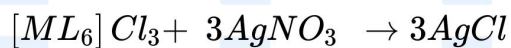
JEE Main 2020 Chapterwise

Chemistry

Q23

$[Ma_3b_3]$ type complex shows facial and meridional isomerism

Q24



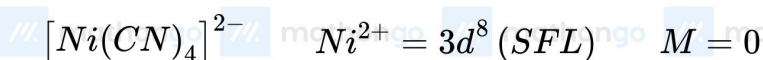
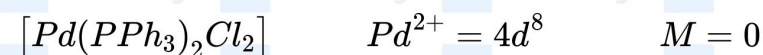
0.3g $\times$ 3 $\times$ 0.125 M

$$\frac{0.3}{267.46} \times 3 = 0.125 \times V \times 10^{-3}$$

$$\text{or, } V = \frac{0.3 \times 3 \times 1000}{267.46 \times 0.125} = 26.92 \text{ ml}$$

Final ans = $26.92 \times 100 = 2692 \text{ ml}$

Q25



So, unpaired electron is 2

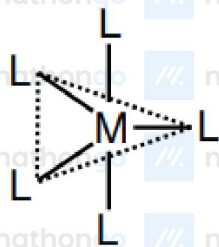
Q26

Ma_2bc can show 2 G.I.

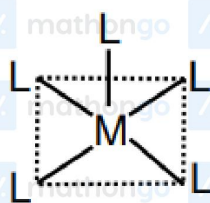
(Square planar)

Ma_4bc can show 2 G.I.

Q27



$$\angle 120^\circ = 3; \angle 90^\circ = 6; \angle 180^\circ = 1 \Rightarrow \text{Total} = 10$$



$$\angle 90^\circ = 8; \angle 180^\circ = 2 \Rightarrow \text{Total} = 10$$

$$\text{Total L - M angles in the two complexes} = 10 + 10 = 20$$

Q28

$$Cr(H_2O)_6Cl_n \left(\mu_{\text{complex}} \right)_{\text{spin}} = 3.8 \text{ B.M.}$$

From data of magnetic moment oxidation number of *Cr* should be +3

Hence complex is $Cr(H_2O)_6Cl_3$

Complex shows geometrical isomerism therefore

formula of complex is $[Cr(H_2O)_4Cl_2] \cdot Cl \cdot 2H_2O$.

It's IUPAC Name: Tetraaquadichloridochromium (III) chloride dihydrate

Q29

Number of Geometrical Isomers in square planar $[PdFCIBrI]^{2-}$ are 3

Hence $n = 3$



$Fe^{3+} = 3d^5$, According to CFT configuration is $t_{2g}^{221} e_g^{00}$

$$\mu = \sqrt{n(n+2)} = 1.73 \text{ B.M.}$$

$$CFSE = -0.4\Delta_0 \times n_{t_{2g}} + 0.6\Delta_0 \times n_{e_g}$$

$$= -0.4\Delta_0 \times 5 = -2.0\Delta_0$$

Q30



Let the O.S. of Fe be x

OS of CN = -1

OS of NOS = -1

$$\therefore x + 5(-1) + (-1) = -4 \Rightarrow x = +2$$



Let O.S. of Fe be y

$$\therefore y + 4(-2) = -4 \Rightarrow y = +4$$



Let O.S. of Fe be z

$$2(z) + 9(0) = 0 \Rightarrow z = 0$$

$$x + y + z = 2 + 4 + 0 = 6$$